PRIEVE ENGINEERING LLC
FORENSIC STRUCTURAL CIVIL

EVALUATION OF CONDITION OF WINDOWS AS RELATES TO HAIL

FILE NUMBER 241008

17756 GEORGE MORAN DRIVE
EDEN PRAIRIE, MN 55347

SITE VISIT DATE: OCTOBER 9, 2024

REPORT DATE: NOVEMBER 4, 2024

I hereby certify that this plan, specification, or report was prepared by me or under my direct supervision and that I am a duly Licensed Professional Engineer under the laws of the state of Minnesota.

Signature: 

Typed or Printed Name: Nathan E Prieve

Date: 11-4-24 License Number: 46203

BACKGROUND:

Per the request of Nic Brau of Farmers Insurance, our office visited the subject property on October 9, 2024. The subject structure is a 2 story, walkout style, wood frame, private residence. The exterior is finished in composite lap siding. The roof membrane is finished in cedar shingles. We understand the structure to have been constructed in 1993. Convective storm activity and hail have been reported on multiple occasions in the area of the subject structure. The roof of the structure was replaced prior to the site visit related to a hail claim. Additional damage has been reported to windows in the home due to hail. Our office was retained to evaluate the condition of windows in the home as relates to hail and to provide recommendations for repairs for purposes of valuation.

This project scope is limited to the specific claim and/or incident at the subject structure. This report has been prepared to assist with evaluation of a property damage claim. This evaluation and report have been prepared within customary and usual standards and budgetary constraints based on information provided to our office. Prieve Engineering LLC is not the Engineer of Record for the subject structure. All other aspects of the subject structure and property are excluded from this evaluation and report.

This report is to be utilized only by the contracted party and is the copyright of Prieve Engineering LLC. This report is only applicable to the subject property and is not to be duplicated or used for any other properties or by any other parties.

ACTIVITIES:

This evaluation included, but was not limited to, review of the following documents and performance of the following activities.

- 2020 Minnesota Residential Code,
- Farmers Insurance Photo Log dated September 11, 2014,
- Farmers Insurance Estimate dated September 16, 2014,
- Farmers Insurance Estimated dated July 15, 2015,
- Inspection Photos dated October 13, 2020,
- Building Loss Worksheet dated 10-7-21,
- Farmers Insurance Letter and Estimate dated February 18, 2022,
- Window Condition Summary (by owner),
- Weather Records,
- Site visit October 9, 2024.

SITE VISIT OCTOBER 9, 2024 OBSERVATIONS:

The site was visited on October 9, 2024. Present front his office was Nathan Prieve P.E. Present at the site were the homeowners. Weather conditions at the time of the site visit were partly cloudy skies with outside temperatures approximately 70 degrees Fahrenheit. For purposes of clarity in the report, the front of the home will be referred to as the west side.

INTERIOR:

The interior of the home was observed at the time of the site visit at areas of reported water entry around windows. Various windows were observed as noted by the owners.

A window group appears at the front/west side above the main entry as shown in photograph A-1 and subsequent photos. A round top center unit appears flanked by adjacent fixed windows. Staining is present on the wall below the left side of the window unit to the upper left corner of the main entry door.

A bay projection appears at the rear/east family room with lower and upper window units along with a high row of single windows. Stain patterns appear below the left side of the single casement facing northeast as shown in photograph A-5. Staining appears below the left side of the main lower center window group in the east wall of the family room. Stains appear at the lower right corner of a window sash as shown n photograph A-7. Additional areas of staining were observed below corners of upper window units as shown in photographs A-9 and A-10. Discoloration was observed along the bottom sill of the lower southeast facing window unit in this area as shown in photograph A-13.

The upper stairway landing windows were observed at the south side, central quadrant of the home. Cracking appears in the wall at the lower left corner of the window unit in the south wall as shown in photograph A-15. The lower stairway landing windows were observed with units in the south and west walls. Cracking appears at the southwest wall corner between the lower corners of windows as shown in photograph A-18.

The window unit in the front wall of the upper southwest bedroom was observed. Apparent staining is present on the wall below the left side of the window. The lower sill of the window was observed to be deteriorated as shown in photograph A-21.

The rear 3 unit window at the upper level northeast bedroom was observed at the time of the site visit with leakage reported from the lower left corner. A penetration was observed in a siding panel below this window at the exterior with impact marks at the center of the penetration. Photographs A-26, A-27 and A-212 show conditions at the penetration in siding.

A four season porch appears at the main level, rear/central quadrant. Wallpaper appears in this room. Apparent staining and discoloration appear in wallpaper below the right corner of the southeast facing window and right side of the northeast facing window.

Windows appear at the northeast corner of the kitchen in the north and east walls. Staining appears at the joint between trim at the lower left corner of the window in the north wall. A southeast facing window appears at the south of the sliding door to the deck in the area south of the kitchen. Patching appears on the wall below the right side of the window as shown in photograph A-37.

EXTERIOR:

Front/West Side;

A triple window unit appears at the main level to the north of the main entry as shown in photograph A-38. Rounded dents were observed at sashes of this window unit. We note that prior claim information and photo logs indicate dents were previously noted at this window unit with replacement of the window unit included with a prior hail claim.

A similar 3 unit window appears at the upper level to the north of the main entry. A scrape mark was observed along the bottom outer frame of the window near a weep. Two dents with sharp marks/indentations appear long the left side of the left sash as shown in photograph A-45. A scrape mark was observed in the face of the bottom corner of a sash not consistent with hail.

The window group above the main entry was observed via at the front entry roof at the time of the site visit. A protrusion was observed along the bottom outer frame of the window unit. Caulking appears at the lower right corner of the window. No hail dents were observed at the window unit.

A triple window group appears at the main level, southwest quadrant on the front. A chip appears in the trim board along the left side of the window bump-out as shown in photograph A-53. We note this chip is visible in prior photo logs from Farmers Insurance dated October 13, 2020. All surfaces of this window were observed with no hail dents visible at frames or sashes.

The upper level bedroom window at the southwest quadrant was observed. Caulk appears at lower corners and lower mullion areas of the window. A dent appears in a window sash with a scrape mark present inside the dent as shown in photograph A-64.

Two single unit windows appear at the west side, north quadrant over the north covered entry door. No hail dents were observed at the sashes or frames of the two windows.

South Side;

Windows were observed at the south side of the home. Two single casement units appear at the west quadrant, one at the main level and one at the upper level. No hail dents were observed at the sashes or frames of the windows.

Two windows appear at the west side of the stairway, one at the upper landing area and one at the lower landing area. No hail dents were observed in frames or sashes of the two windows at the west side of the stairway.

The lower main stairway window group was observed at the south side, central quadrant. No hail dents were observed at sashes or frames of this window group.

The main upper stairway windows were observed via a ladder. Caulking appears over mullions of the upper window section along with lower corners of window frames. All surfaces were observed including sashes and frames with no hail dents visible on windows. A chip was observed in the trim board at the wall corner to the east of the stairway with underlying surfaces aged as shown in photograph A-90.

East Side;

Three levels appear exposed at the rear/east side of the home. Lower level windows and a patio door were observed under the deck and porch. No dents or marks were observed at windows and doors under the deck. A 3 unit window appears at the lower level, northeast quadrant of the home. No hail dents were observed at this window unit at the frame or sashes.

The main level family room windows were observed at the southeast quadrant as shown in photograph A-101 and subsequent photos. A dent with scrape marks was observed in the frame of the upper southeast facing window as shown in photograph A-107. Caulk appears at various areas of windows including corners of sashes and at mullions and corners of frames. An indentation appears on the left side of the center window accessory groove not consistent with hail along with a mark in the frame with loss of finishes as shown in photograph A-113. Two dents appear in the face of the sash at the lower left corner of the main window with scrape marks present inside and around areas of dents. A dent with marks present appears in the face of the drip cap of the upper center window unit. A scrape mark appears in the face of the frame of this window as shown in photograph A-121. A dent appears in the bottom accessory groove of the upper northeast facing window not consistent with hail. An upper bank of single window units appears above the bay projection at the south quadrant. Caulk appears at lower corners of frames. No hail dents appear at sashes or frames at the single unit windows. A scrape mark appears in the sash of the north window unit as shown in photograph A-146.

Two sliding doors appear at the rear deck with one facing east and one facing south at the porch. A southeast facing window appears to the left of the doors. Scrape marks appear in the frame of the window along the right side not consistent with hail. A dent with a scrape mark present appears in the bottom screen frame of the east facing sliding door.

A bay projection appears at the upper level, central quadrant above the east deck door. A dent appears along the left accessory groove of the southeast facing window not consistent with hail. Various scrape marks and a dent appear along the bottom frame of the center window not consistent with hail. Dents with scrape marks appear at the lower left corner of the sash of the northeast facing window not consistent with hail.

An upper level east facing window appears over the four season porch at the rear side. A dent appears along the bottom accessory groove not consistent with hail. A similar dent appears along the left side as shown in photograph A-163 not consistent with hail.

Windows were observed around the four season porch with southeast, east, northeast and north facing windows present. Scrape marks and dents with sharp marks appear at the southeast facing window as shown in photographs A-165 and A-166. Caulk appears visible along the bottom of the east facing window against glazing. The drip cap at this window was observed uneven at the right portion. A dent was observed along the bottom accessory groove of the northeast facing window as shown in photograph A-174. A four unit window appears in the north wall of the porch. Two dents appear at the lower left corner of the left casement which are generally smooth. This window was noted to have been included for full replacement from a prior hail claim as noted in past estimates provided by Farmers Insurance. A dent appears in the bottom accessory groove of the window not consistent with hail.

The north and east facing windows at the kitchen were observed. A scrape mark and dent appear at the right side of the east facing window not consistent with hail. Caulk appears at

the lower right corners of the north kitchen window. No hail dents appear at sashes or frames of kitchen windows.

A 3 unit window appears at the main level, northeast quadrant of the home. A dent appears in the left accessory groove similar to those observed elsewhere. Scrape marks appear along the lower frame as shown in photograph A-199 and A-201. Caulk appears at lower mullions of the window.

An upper level north facing window was observed above the kitchen. No hail dents were observed at the window sash or frame. A dent appears in the left accessory groove at the lower corner of the window.

The upper level 3 unit bedroom window was observed from a ladder at the northeast quadrant. No hail marks or dents were observed at this window at the frame or sashes.

A 2 unit window appears at the rear/east side of the garage adjacent to a shed. Various scrape marks appear in the frame of the window not consistent with hail. No smooth or rounded dents were observed in the frame or sashes.

Weather Records;

Weather data was reviewed for the area of the subject structure including stormersite. Reported observed past hail in the area includes 1.25” on June 19, 2015 to the east of the subject structure, 1.75” on July 25, 2017 to the east of the structure, 1” hail on September 6, 2020 to the east of the property, and multiple reports on August 9, 2020 ranging from 1 to 2 inches to the east, west and north of the property. Similar past estimated hail appears in a summary from the NOAA Storm Events Database with most recent hail in Eden Prairie reported in 2020.

CONCLUSIONS AND RECOMMENDATIONS:

After visiting the site, reviewing available data and documents, and based upon the training and expertise of the investigator, we have formed the following conclusions and recommendations to a reasonable degree of engineering certainty.

Prieve Engineering LLC reserves the right to supplement or amend the report pending further discovery. Any and all recommendations in the report are for budgeting and estimating purposes only and are not for permitting and/or construction. Detailed recommendations for permitting and construction will require separate stamped drawings and contract. Prieve Engineering LLC is not the Engineer of Record for the subject structure.

1) It is our opinion that no recent hail damage has occurred to windows in the subject structure related to any recent hail or other storm activity. Past storm data indicates a storm occurred on or around August 9, 2020 in the area. This is consistent with past claims including replacement of the roof due to hail damage. We note that prior claim estimates and information include 2 windows in the home noted for potential hail damage and replacement including the main level window to the left of the main entry and north facing window at the rear four season porch. Dents shown in prior claim photographs are similar to those observed during the site inspection. Given these 2 windows have already been included and we understand paid for due to prior storm activity, no additional payments in our opinion are appropriate for the 2 windows.

2) It is our opinion that no other damage occurred to windows around the home due to hail or any other catastrophic causation that warrants any repair or replacement. Dents and scrape marks appear at various areas of windows around the home not constant with hail. Dents noted at sashes and frames in our opinion include scrape marks inside dents consistent with impacts from blunt objects other than hail. Dents in accessory grooves are consistent with impacts other than hail or damage occurring during or following installation of windows at the time of construction of the home. Further, it is our opinion that past hail at the site was not of sufficient size to dent window frames or the face of sashes. No repairs or replacement of any further windows is warranted or required at the subject property due to hail or any other catastrophic causation other than the 2 windows already included and paid in prior claims.

3) It is our opinion that reported water leakage and water entry at windows is unrelated to hail or any other catastrophic causation. Stains present on walls at certain windows in our opinion are consistent with condensation and/or leakage through windows. We note that many windows have various applications of caulking at exterior sashes and frames consistent with past attempts to prevent water entry. Areas of staining and deterioration appear at the interior of windows consistent with long term aging and deterioration to units. The conditions observed at windows, including reported leakage and water entry, in our opinion relate to wear and tear and aging of units and are unrelated to hail or any other catastrophic causation. No repairs or replacement of windows are warranted as relates to hail and alleged leakage of units.

4) It is our opinion that the siding penetration at the rear upper level of the home is unrelated to hail. Marks appear inside the penetration consistent with an impact from a blunt object other than hail. Further, recent storm activity in the area generally consists of storms moving from west to east with hail originating from a westerly direction. The penetration in siding appears at the east side, opposite typical origination of hail in recent storm activity in the area. No repairs in our opinion are warranted to siding as relates to hail.